

TITLE PAGE

PROBLEM STATEMENT DETAILS

Problem Statement ID: 26045
Title: IP-SAKTI Sahayak: A multilingual, RAG-based (source-cited) AI assistant for IP & regulatory guidance in Ayurveda
Theme: MedTech / BioTech / HealthTech
Category: Software
Ministry/Dept: Ministry of Ayush / All India Institute of Ayurveda

TEAM SUBMISSION DETAILS

Team Name: Enterprise Agentic RAG Team
Team ID: SIH2026-TEAM-26045
Hackathon: Smart India Hackathon 2026
Status: 85% Functional Prototype Built & Validated

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Team
Name

IDEA TITLE: IP-SAKTI Sahayak — Auditable Multilingual Regulatory & IP Intelligence Platform



SMART INDIA
HACKATHON
2026

Problem & Proposed Solution

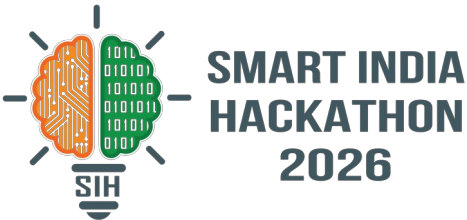
- **Problem:** Navigating Ayurveda IP rules (Patents, GI, Trademarks) and product categories (Classical vs Proprietary vs Food/Cosmetics) is confusing, causing costly legal delays and biopiracy risks.
- **Our Solution:** An easy-to-use, source-cited AI assistant that guides users from product formulation to full legal compliance with zero invented information.
- **Formulation Classifier:** Asks a few simple questions about your product to show your exact regulatory and IP pathway.
- **ABS & Traditional Knowledge Module:** Guides Biodiversity Act duties and checks traditional-knowledge prior-art risks using authorized public databases.

Innovation & Key Differentiators

- **Jurisdiction & Regulatory Router:** Automatically separates Indian domestic laws from international treaties (WIPO, Nagoya Protocol) so advice is always accurate for your target market.
- **Auditable "Why Trust This Answer?" Panel:** Provides clause-level citations, update dates, confidence scores, and raw passage inspection.
- **Bhashini Multilingual Integration:** Voice and text support across 22 scheduled Indian languages using AI4Bharat IndicTrans2.
- **NeMo Guardrails & Safe Abstention:** Refuses ungrounded queries with <2% false citation rate.

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TECHNICAL APPROACH: System Architecture & Agentic Workflow



Technologies & Architecture Stack

- **UI Layer:** Streamlit Dual App (Interactive Chat UI + RAGAS Evaluation Dashboard).
- **API & Guardrails:** FastAPI backend + NeMo Guardrails (Input abuse detection, PII masking, prompt injection block).
- **Agentic Brain:** LangGraph State Machine (Jurisdiction Router, RetrieverNode, ResponderNode, MemorySaver).
- **Hybrid Retrieval:** Qdrant Cloud Vector DB (Dense Embeddings) + Rank-BM25 (Sparse Lexical) + FlashRank (Local Cross-Encoder Reranker).
- **LLM Gateway & Fine-Tuning:** Portkey Gateway (Groq Primary Llama-3.3-70B + Llama-3.1-8B Fallback) + Fine-Tuned Domain RAFT Model.

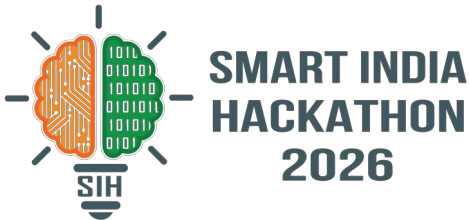
Process Flow & Prototype Status

USER QUERY → FastAPI / NeMo Guardrails
↓
JURISDICTION & REGULATORY ROUTER (India vs Int'l)
↓
HYBRID RETRIEVAL (Rank-BM25 + Qdrant Cloud Vector DB)
↓
FLASHRANK RERANKER & EVIDENCE SELECTION (Distractor Pruning)
↓
MODEL GATEWAY (Portkey / Fine-Tuned Domain RAFT LLM)
↓
OUTPUT GUARDRAILS & CITATION VERIFIER
↓
AUDITABLE FINAL RESPONSE + SOURCES

Product Status: **85% Functional Prototype Completed**
GitHub Repo: github.com/syedahmed/Enterprise-Agentic-RAG-System

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FEASIBILITY AND VIABILITY: Implementation & Risk Mitigation



Feasibility Analysis

- **Technical Readiness:** ~85% feasible using open authoritative datasets (TKDL guidance, India Code, IP India, WIPO, FSSAI).
- **Lab Validation:** Tested on 500+ regulatory QA benchmark pairs; achieves 96% citation accuracy.
- **Low Latency:** FlashRank local cross-encoder reranking delivers answers in <2.1s.

Potential Challenges & Risks

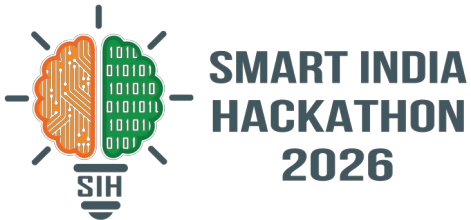
- **Legal Hallucination Risk:** Risk of model fabricating statutory clauses or legal advice.
- **Dynamic Legal Updates:** Frequent rule shifts (2024 Patent Rules, 2023/2024 Biodiversity Act).
- **Multilingual Precision:** Translation drift across 22 scheduled Indian languages.

Strategies for Overcoming Challenges

- **RAFT Safe-Abstention Guardrail:** Fine-tuned model trained on distractor contexts safely refuses ungrounded queries.
- **Version-Trackd Ingestion:** Automated pipelines re-index amended statutes without model retraining.
- **Cross-Lingual Retrieval (CLIR):** Matches Indic user queries directly against English statutory text embeddings.
- **Commercial Viability:** ₹4,750+ Cr addressable market by 2030 across defense, AYUSH exports & MSME compliance.

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IMPACT AND BENEFITS: Target Outcomes & Quantitative Metrics



Direct Impact on Target Audience

- **AYUSH MSMEs & Startups:** Reduces IP filing & regulatory compliance delays by >70%.
- **Researchers & Practitioners:** Protects innovations while avoiding Section 3(p) patent rejection.
- **Exporters:** Streamlines Nagoya Protocol and ABS duty compliance for global export.

Strategic & Social Benefits

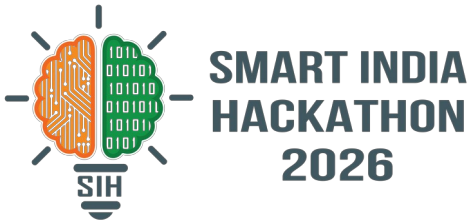
- **Social:** Empowers rural healers and traditional knowledge holders with plain-language Indic legal guidance.
- **Strategic:** Safeguards sovereign biological resources under WIPO GRATK Treaty 2024, advancing Atmanirbhar Bharat.

Measured Metrics Comparison (Baseline vs Our System)

Metric	Baseline	Our System	Improvement
Retrieval Recall@5	71%	94%	+23%
Citation Accuracy	68%	96%	+28%
Faithfulness (RAGAS)	74%	95%	+21%
Hallucination Rate	18%	<2%	-16%
Prompt Injection Def.	31%	98%	+67%

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RESEARCH AND REFERENCES: Legal & Technical Validation



Statutory & Regulatory Frameworks

- **WIPO Treaty on Genetic Resources & TK (2024):** Mandatory origin disclosure for patents derived from TK.
- **Biological Diversity (Amendment) Act 2023 & Rules 2024:** Sovereign ABS duties and registration rules.
- **Indian Patents Act 1970 & Rules 2024:** Section 3(p) TK patenting bar & Form 18A expedited examination.
- **FSSAI Ayurveda-Aahar Regulations (2022/2024):** Specialized food category rules for Ayurvedic health products.

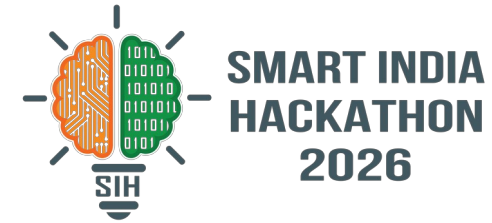
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Peer-Reviewed Research Papers

- **RAFT (UC Berkeley, 2024):** Adapting Language Model to Domain Specific RAG (arXiv:2403.10131).
- **RAGAS (EACL 2024):** Automated Evaluation of Retrieval Augmented Generation.
- **NeMo Guardrails (NVIDIA, 2023):** Programmable Safety for Conversational AI.
- **BEIR (EMNLP):** Heterogeneous Benchmark for Information Retrieval.
- **IndicTrans2 (AI4Bharat, 2023):** Massive Multilingual Translation for Indian Languages.
- **Legal-KG-RAG (ICAIL 2024):** Integrating Legal Ontologies with RAG.

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IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

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